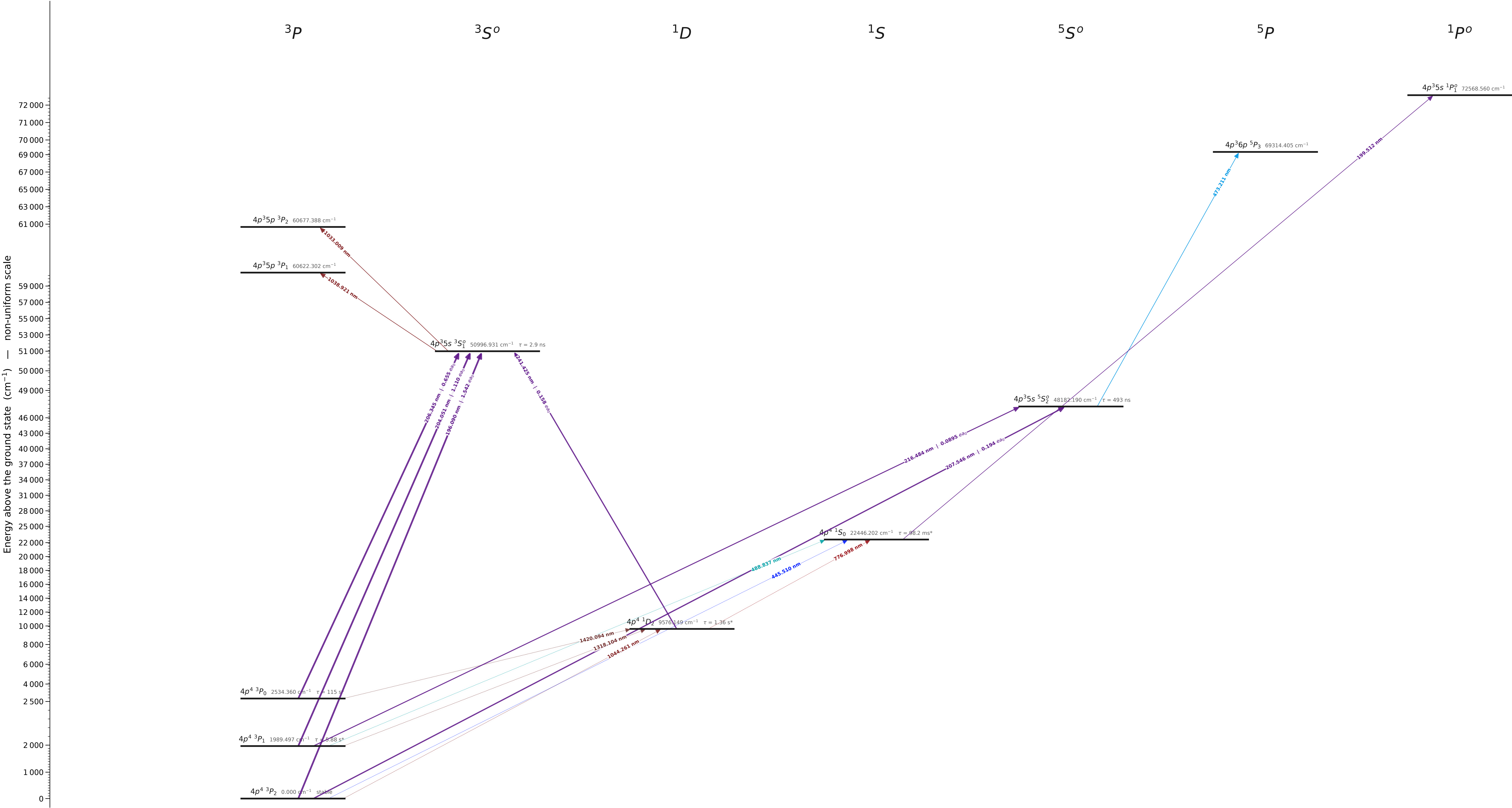
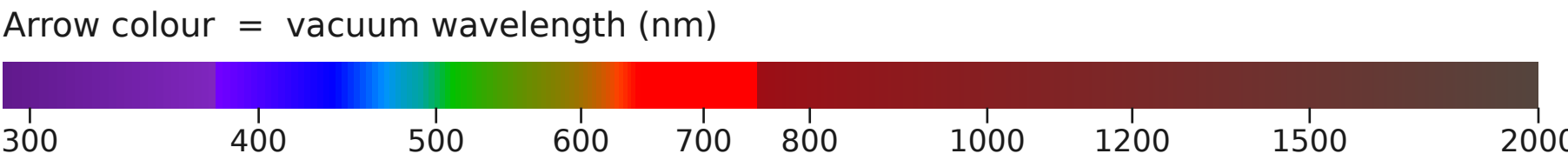


# Selenium

Se (Se I). Strongest classified transitions below 2 μm (NIST): 11 levels, 16 lines.

Se<sup>+</sup> ionisation limit 78658.15 cm<sup>-1</sup> = 9.75237 eV (127.132 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $| \langle J \| er \| J' \rangle |$  in  $ea_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                            | λ vac (nm) | λ air (nm) | ν (THz)   | ⟨J  er  J'⟩  (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                             |
|---------------------------------------|------------|------------|-----------|---------------------------------|----------------------|----------------|------------|-------------------------------------------------|
| $4p^4\ ^3P_2 - 4p^35s\ ^3S_1^\circ$   | 196.0902   | -          | 1528.8495 | 1.542                           | 2.130e+08            | 54.88          | 61.7       | resonance line from the ground term             |
| $4p^4\ ^1S_0 - 4p^35s\ ^1P_1^\circ$   | 199.5118   | -          | 1502.6305 | -                               | -                    | -              | -          | classified strong                               |
| $4p^4\ ^3P_1 - 4p^35s\ ^3S_1^\circ$   | 204.0507   | 203.9852   | 1469.2059 | 1.110                           | 9.800e+07            | 54.88          | 28.4       | resonance line from the ground term             |
| $4p^4\ ^3P_0 - 4p^35s\ ^3S_1^\circ$   | 206.3448   | 206.2789   | 1452.8713 | 0.655                           | 3.300e+07            | 54.88          | 9.52       | resonance line from the ground term             |
| $4p^4\ ^3P_2 - 4p^35s\ ^5S_2^\circ$   | 207.5456   | 207.4794   | 1444.4657 | 0.194                           | 1.700e+06            | 0.3228         | 84         | 207.5 nm intercombination line                  |
| $4p^4\ ^3P_1 - 4p^35s\ ^5S_2^\circ$   | 216.4845   | 216.4165   | 1384.8221 | 0.090                           | 3.200e+05            | 0.3228         | 16         | 216.4 nm intercombination line                  |
| $4p^4\ ^1D_2 - 4p^35s\ ^3S_1^\circ$   | 241.4247   | 241.3513   | 1241.7638 | 0.158                           | 1.200e+06            | 54.88          | 0.34       | weak branch of 3S°1 to the metastable 1D2 level |
| $4p^4\ ^3P_2 - 4p^4\ ^1S_0$           | 445.5097   | 445.3846   | 672.9202  | -                               | 1.800e-01            | 1.621e-06      | 1.77       | forbidden line within the ground configuration  |
| $4p^35s\ ^3S_2^\circ - 4p^36p\ ^5P_3$ | 473.2112   | 473.0788   | 633.5279  | -                               | -                    | -              | -          | classified strong                               |
| $4p^4\ ^3P_1 - 4p^4\ ^1S_0$           | 488.8373   | 488.7008   | 613.2766  | -                               | 7.700e+00            | 1.621e-06      | 75.6       | forbidden line within the ground configuration  |
| $4p^4\ ^1D_2 - 4p^4\ ^1S_0$           | 776.9976   | 776.7838   | 385.8345  | -                               | 2.300e+00            | 1.621e-06      | 22.6       | forbidden line within the ground configuration  |
| $4p^35s\ ^3S_1^\circ - 4p^35p\ ^3P_2$ | 1033.0091  | 1032.7260  | 290.2128  | -                               | -                    | -              | -          | classified strong                               |
| $4p^35s\ ^3S_1^\circ - 4p^35p\ ^3P_1$ | 1038.9210  | 1038.6364  | 288.5614  | -                               | -                    | -              | -          | classified strong                               |
| $4p^4\ ^3P_2 - 4p^4\ ^1D_2$           | 1044.2611  | 1043.9750  | 287.0857  | -                               | 6.270e-01            | 1.17e-07       | 85.3       | forbidden line within the ground configuration  |
| $4p^4\ ^3P_1 - 4p^4\ ^1D_2$           | 1318.1045  | 1317.7441  | 227.4421  | -                               | 1.100e-01            | 1.17e-07       | 15         | forbidden line within the ground configuration  |
| $4p^4\ ^3P_0 - 4p^4\ ^1D_2$           | 1420.0937  | 1419.7055  | 211.1075  | -                               | 9.200e-05            | 1.17e-07       | 0.01       | forbidden line within the ground configuration  |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).